Администрирование локальных сетей

Динамическая маршрутизация OSPF

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Цели и задачи работы

Цель лабораторной работы

Настроить динамическую маршрутизацию между территориями организации и проверить изменение маршрутов при отказе одного из каналов связи.

Задачи лабораторной работы

- настроить OSPF на маршрутизаторах организации;
- организовать прямую связь сети квартала 42 в Москве с филиалом в Сочи;
- проверить соседства OSPF и таблицы маршрутизации;
- отследить движение ICMP-пакетов в режиме Simulation;
- проверить изменение маршрута при отключении и восстановлении VLAN 6.

Исходная топология сети

Общая схема сети

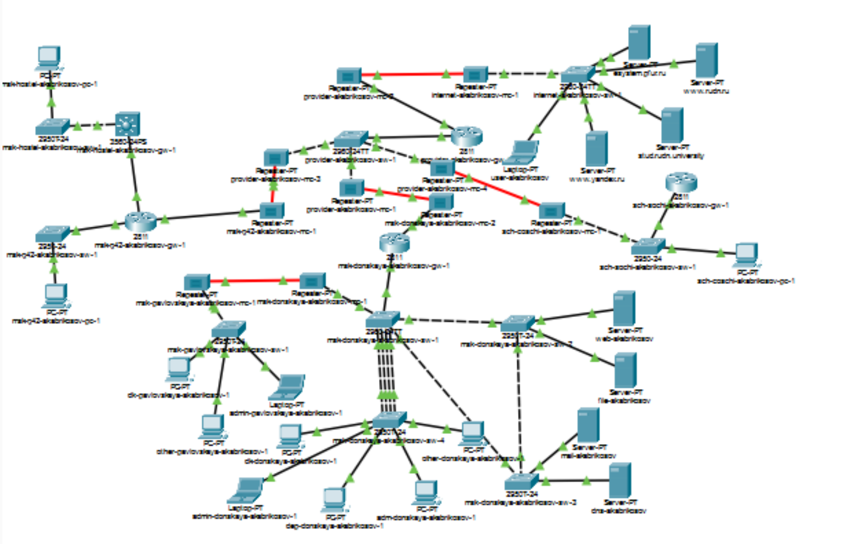

Рис. 1: Общая топология сети организации

Настройка динамической маршрутизации OSPF

```
Password:
msk-donskaya-akabrikosov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-akabrikosov-gw-1(config)#
msk-donskaya-akabrikosov-gw-1(config)#router ospf 1
msk-donskaya-akabrikosov-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-akabrikosov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-donskaya-akabrikosov-gw-1(config-router)#exit
msk-donskaya-akabrikosov-gw-1(config)#exit
msk-donskaya-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-akabrikosov-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-akabrikosov-gw-1#
```

Рис. 2: Настройка OSPF на маршрутизаторе msk-donskaya-gw-1

Настройка OSPF на msk-q42-gw-1

```
msk-q42-akabrikosov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-akabrikosov-gw-1(config)#
msk-q42-akabrikosov-gw-1(config)#router ospf 1
msk-q42-akabrikosov-gw-1(config-router)#router-id 10.128.254.2
msk-q42-akabrikosov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-q42-akabrikosov-gw-1(config-router)#exit
msk-q42-akabrikosov-gw-1(config)#exit
msk-q42-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-akabrikosov-gw-1#
00:52:40: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/1.5 from LOADING to FULL, Loading
Done

msk-q42-akabrikosov-gw-1#int f0/1.7
                                 ^
% Invalid input detected at '^' marker.

msk-q42-akabrikosov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-akabrikosov-gw-1(config)#int f0/1.7
msk-q42-akabrikosov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-akabrikosov-gw-1(config-subif)#enc dot1q 7
msk-q42-akabrikosov-gw-1(config-subif)#ip addr 10.128.255.9 255.255.255.252
msk-q42-akabrikosov-gw-1(config-subif)#descr sochi
msk-q42-akabrikosov-gw-1(config-subif)#exit
msk-q42-akabrikosov-gw-1(config)#exit
msk-q42-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Рис. 3: Настройка OSPF и подынтерфейса VLAN 7 на msk-q42-gw-1

Настройка OSPF на msk-hostel-gw-1

```
msk-hostel-akabrikosov-gw-1>en
Password:
msk-hostel-akabrikosov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-akabrikosov-gw-1(config)#
msk-hostel-akabrikosov-gw-1(config)#router ospf 1
msk-hostel-akabrikosov-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-akabrikosov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-hostel-akabrikosov-gw-1(config-router)#exit
msk-hostel-akabrikosov-gw-1(config)#exit
msk-hostel-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-akabrikosov-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-akabrikosov-gw-1#
00:55:56: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL, Loading Done

msk-hostel-akabrikosov-gw-1#
```

Рис. 4: Настройка OSPF на маршрутизаторе msk-hostel-gw-1

Настройка OSPF на sch-sochi-gw-1

```
sch-sochi-akabrikosov-gw-1>en
Password:
sch-sochi-akabrikosov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-akabrikosov-gw-1(config)#router ospf 1
sch-sochi-akabrikosov-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-akabrikosov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
sch-sochi-akabrikosov-gw-1(config-router)#exit
sch-sochi-akabrikosov-gw-1(config)#
00:54:30: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to FULL, Loading
Done

sch-sochi-akabrikosov-gw-1(config)#int f0/0.7
sch-sochi-akabrikosov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-akabrikosov-gw-1(config-subif)#enc dot1q 7
sch-sochi-akabrikosov-gw-1(config-subif)#ip addr 10.128.255.10 255.255.255.252
sch-sochi-akabrikosov-gw-1(config-subif)#descr q42
sch-sochi-akabrikosov-gw-1(config-subif)#exit
sch-sochi-akabrikosov-gw-1(config)#exit
sch-sochi-akabrikosov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-akabrikosov-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-akabrikosov-gw-1#
```

Рис. 5: Настройка OSPF и подынтерфейса VLAN 7 на sch-sochi-gw-1

Настройка прямой связи Москва—Сочи

Создание VLAN 7 на коммутаторе провайдера

```
provider-akabrikosov-sw-1>en
Password:
provider-akabrikosov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-akabrikosov-sw-1(config)#vlan 7
provider-akabrikosov-sw-1(config-vlan)#name q42-sochi
provider-akabrikosov-sw-1(config-vlan)#int vlan 7
provider-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-akabrikosov-sw-1(config-if)#no sh
provider-akabrikosov-sw-1(config-if)#exit
provider-akabrikosov-sw-1(config)#exit
provider-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
provider-akabrikosov-sw-1#
```

Рис. 6: Создание VLAN 7 на коммутаторе provider-akabrikosov-sw-1

Создание VLAN 7 на коммутаторе филиала в Сочи

```
sch-sochi-akabrikosov-sw-1>en
Password:
sch-sochi-akabrikosov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-akabrikosov-sw-1(config)#
sch-sochi-akabrikosov-sw-1(config)#vlan 7
sch-sochi-akabrikosov-sw-1(config-vlan)#name q42-sochi
sch-sochi-akabrikosov-sw-1(config-vlan)#int vlan 7
sch-sochi-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-akabrikosov-sw-1(config-if)#no sh
sch-sochi-akabrikosov-sw-1(config-if)#exit
sch-sochi-akabrikosov-sw-1(config)#exit
sch-sochi-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-akabrikosov-sw-1#
```

Рис. 7: Создание VLAN 7 на коммутаторе sch-sochi-akabrikosov-sw-1

Проверка работы OSPF

Проверка процесса OSPF и соседей

```
msk-donskaya-akabrikosov-gw-1#sh ip ospf
 Routing Process "ospf 1" with ID 10.128.254.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 8
        Area has no authentication
        SPF algorithm executed 7 times
        Area ranges are
        Number of LSA 7. Checksum Sum 0x041fdb
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0

msk-donskaya-akabrikosov-gw-1#
msk-donskaya-akabrikosov-gw-1#sh ip ospf nei


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.128.254.2      1   FULL/BDR        00:00:38    10.128.255.2    FastEthernet0/1.5
10.128.254.4      1   FULL/BDR        00:00:30    10.128.255.6    FastEthernet0/1.6
msk-donskaya-akabrikosov-gw-1#
```

Рис. 8: Проверка процесса OSPF и соседей на msk-donskaya-gw-1

Таблица маршрутизации msk-donskaya-gw-1

```
Gateway of last resort is 198.51.100.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 24 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:00:23, FastEthernet0/1.5
                        [110/2] via 10.128.255.6, 00:00:23, FastEthernet0/1.6
S       10.129.0.0/16 [1/0] via 10.128.255.2
O       10.129.0.0/24 [110/2] via 10.128.255.2, 00:06:21, FastEthernet0/1.5
O       10.129.1.0/24 [110/2] via 10.128.255.2, 00:04:29, FastEthernet0/1.5
O       10.129.128.0/24 [110/3] via 10.128.255.2, 00:04:19, FastEthernet0/1.5
S       10.130.0.0/16 [1/0] via 10.128.255.6
O       10.130.0.0/24 [110/2] via 10.128.255.6, 00:03:21, FastEthernet0/1.6
O       10.130.1.0/24 [110/2] via 10.128.255.6, 00:03:21, FastEthernet0/1.6
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-akabrikosov-gw-1#
```

Проверка маршрутов на msk-q42-gw-1

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.128.254.4      1   FULL/DR         00:00:31    10.128.255.10   FastEthernet0/1.7
10.128.254.1      1   FULL/DR         00:00:39    10.128.255.1    FastEthernet0/1.5
10.128.254.3      1   FULL/BDR        00:00:30    10.129.1.2      FastEthernet1/0.202
msk-q42-akabrikosov-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O       10.128.0.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
O       10.128.1.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
O       10.128.3.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
O       10.128.4.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
O       10.128.5.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
O       10.128.6.0/24 [110/2] via 10.128.255.1, 00:07:07, FastEthernet0/1.5
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
O       10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:14, FastEthernet0/1.5
                        [110/2] via 10.128.255.10, 00:01:14, FastEthernet0/1.7
C       10.128.255.8/30 is directly connected, FastEthernet0/1.7
L       10.128.255.9/32 is directly connected, FastEthernet0/1.7
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
O       10.129.128.0/24 [110/2] via 10.129.1.2, 00:05:15, FastEthernet1/0.202
O       10.130.0.0/24 [110/2] via 10.128.255.10, 00:01:14, FastEthernet0/1.7
O       10.130.1.0/24 [110/2] via 10.128.255.10, 00:01:14, FastEthernet0/1.7
S*   0.0.0.0/0 [1/0] via 10.128.255.1

msk-q42-akabrikosov-gw-1#
```

Проверка маршрутов на msk-hostel-gw-1

```
msk-hostel-akabrikosov-gw-1#sh ip ospf nei


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.128.254.2      1   FULL/DR         00:00:34    10.129.1.1      Vlan202
msk-hostel-akabrikosov-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O       10.128.0.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.1.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.3.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.4.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.5.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.6.0/24 [110/3] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.255.0/30 [110/2] via 10.129.1.1, 00:05:47, Vlan202
O       10.128.255.4/30 [110/3] via 10.129.1.1, 00:04:37, Vlan202
O       10.128.255.8/30 [110/2] via 10.129.1.1, 00:01:49, Vlan202
O       10.129.0.0/24 [110/2] via 10.129.1.1, 00:05:47, Vlan202
C       10.129.1.0/24 is directly connected, Vlan202
C       10.129.128.0/24 is directly connected, Vlan301
O       10.130.0.0/24 [110/3] via 10.129.1.1, 00:01:49, Vlan202
O       10.130.1.0/24 [110/3] via 10.129.1.1, 00:01:49, Vlan202
S*   0.0.0.0/0 [1/0] via 10.129.1.1

msk-hostel-akabrikosov-gw-1#
```

Проверка маршрутов на sch-sochi-gw-1

```
sch-sochi-akabrikosov-gw-1#sh ip ospf nei


Neighbor ID     Pri   State           Dead Time   Address         Interface
10.128.254.1      1   FULL/DR         00:00:37    10.128.255.5    FastEthernet0/0.6
10.128.254.2      1   FULL/BDR        00:00:30    10.128.255.9    FastEthernet0/0.7
sch-sochi-akabrikosov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 18 subnets, 3 masks
O       10.128.0.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.1.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.3.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.4.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.5.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.6.0/24 [110/2] via 10.128.255.5, 00:05:01, FastEthernet0/0.6
O       10.128.255.0/30 [110/2] via 10.128.255.5, 00:02:13, FastEthernet0/0.6
                        [110/2] via 10.128.255.9, 00:02:13, FastEthernet0/0.7
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.128.255.8/30 is directly connected, FastEthernet0/0.7
L       10.128.255.10/32 is directly connected, FastEthernet0/0.7
O       10.129.0.0/24 [110/2] via 10.128.255.9, 00:02:13, FastEthernet0/0.7
O       10.129.1.0/24 [110/2] via 10.128.255.9, 00:02:13, FastEthernet0/0.7
O       10.129.128.0/24 [110/3] via 10.128.255.9, 00:02:13, FastEthernet0/0.7
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*   0.0.0.0/0 [1/0] via 10.128.255.5

sch-sochi-akabrikosov-gw-1#
```

Анализ прохождения ICMP-пакетов

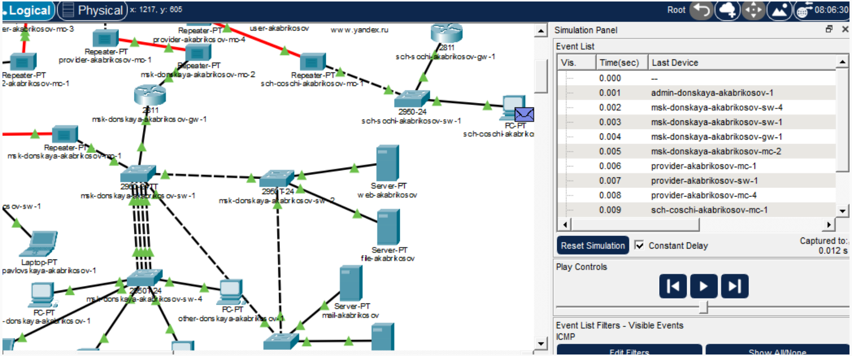

Рис. 13: Передача ICMP-пакета от ноутбука администратора до компьютера в Сочи

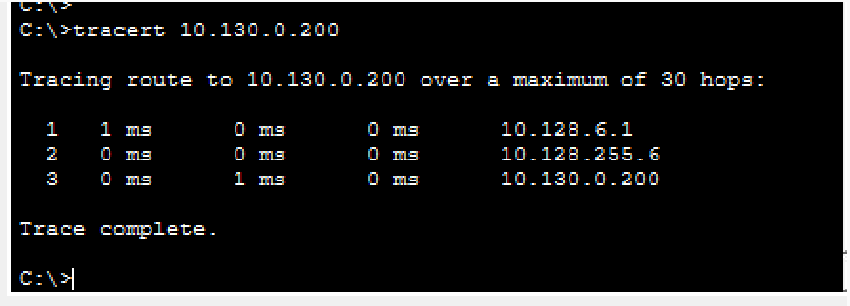

Рис. 14: Трассировка маршрута до компьютера пользователя в Сочи при включенном VLAN 6

Проверка отказоустойчивости маршрутизации

```
provider-akabrikosov-sw-1(config)#
provider-akabrikosov-sw-1(config)#int vlan 6
provider-akabrikosov-sw-1(config-if)#sh

provider-akabrikosov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down

provider-akabrikosov-sw-1(config-if)#exit
provider-akabrikosov-sw-1(config)#no vlan 6
provider-akabrikosov-sw-1(config)#exit
provider-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-akabrikosov-sw-1#wr mem
Building configuration...
[OK]
provider-akabrikosov-sw-1#
```

Рис. 15: Временное отключение VLAN 6 на коммутаторе провайдера

Изменение маршрута после отключения VLAN 6

```
C:\>
C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.128.6.1
  2   0 ms      0 ms      1 ms      10.128.255.2
  3   0 ms      0 ms      0 ms      10.128.255.10
  4   0 ms      0 ms      0 ms      10.130.0.200

Trace complete.
```

Рис. 16: Изменение маршрута до Сочи после отключения VLAN 6

Восстановление VLAN 6

```
provider-akabrikosov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-akabrikosov-sw-1(config)#vlan 6
provider-akabrikosov-sw-1(config-vlan)#name sochi
provider-akabrikosov-sw-1(config-vlan)#int vlan 6
provider-akabrikosov-sw-1(config-if)#no sh

provider-akabrikosov-sw-1(config-if)#
%LINK-3-UPDOWN: Interface Vlan6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-akabrikosov-sw-1(config-if)#exit
provider-akabrikosov-sw-1(config)#exit
provider-akabrikosov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-akabrikosov-sw-1#wr mem
```

Рис. 17: Восстановление VLAN 6 на коммутаторе провайдера

Возврат основного маршрута

```
C:\>
C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      10.128.6.1
  2   2 ms      0 ms      0 ms      10.128.255.6
  3   0 ms      0 ms      0 ms      10.130.0.200

Trace complete.

C:\>
```

Рис. 18: Трассировка маршрута после восстановления VLAN 6

Повторная проверка ICMP

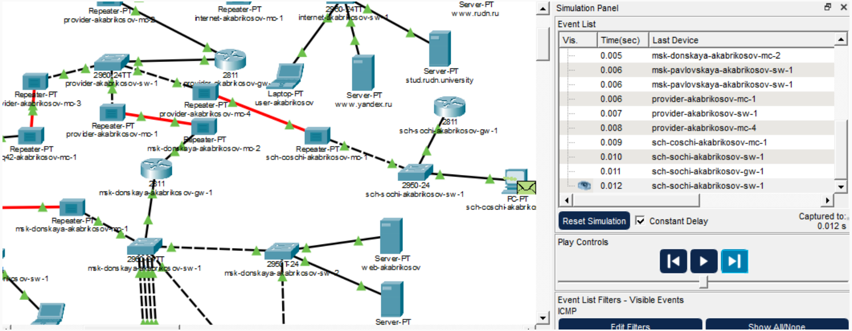

Рис. 19: Проверка прохождения ICMP-пакета после восстановления VLAN 6

Выводы по проделанной работе

Вывод

В ходе лабораторной работы была настроена динамическая маршрутизация по протоколу OSPF между маршрутизаторами московских площадок и филиала в Сочи.

Были сформированы OSPF-соседства, проверены таблицы маршрутизации и настроена прямая связь между кварталом 42 и филиалом в Сочи через VLAN 7.

При отключении VLAN 6 маршрут до филиала в Сочи изменился: трафик стал проходить через сеть квартала 42 по альтернативному OSPF-маршруту.

После восстановления VLAN 6 трафик вернулся на основной путь, что подтвердило автоматическую перестройку маршрутов средствами OSPF.